

Comparing the Decaying Turbulence Theory with Wind Tunnel Experiments

Alexander Migdal^{1,*}

¹*Institute for Advanced Study, Princeton, NJ, USA*

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Abstract

The Euler-ensemble solution of freely decaying incompressible turbulence predicts a universal, parameter-free second-order structure function as a Mellin–Barnes integral in the scaling variable $\rho = r/\sqrt{\tilde{\nu}(t+t_0)}$. We evaluate this integral for the odd Euler ensemble, the only marginally stable turbulent attractor, by steepest descent along Lefschetz thimbles with Gauss–Hermite quadrature, including the Stokes phenomenon: the residues of the Riemann-wall and dyadic-wall poles trapped by the thimble, and the thimbles that terminate on the paired zeros. The computation agrees with a direct contour integration to 5×10^{-10} . The same Riemann-wall and dyadic-wall poles that produce the log-periodic oscillations of the energy spectrum also produce log-periodic oscillations of the effective index $\alpha(\rho)$. They appear only at small separations, $\rho < C_{\min}$, with an amplitude below 1.5×10^{-6} , eight to nine orders of magnitude smaller than in the spectrum. For $\rho > C_{\max}$ there are no oscillations at all. Finite-size effects are included exactly by subtracting the infrared part $k < \pi/W$. We compare the theory with the Max Planck wind-tunnel measurements of Küchler, Bewley and Bodenschatz at $Re_\lambda = 413\text{--}5779$. The small oscillations of the measured index beyond the tunnel width, $r > W$, are reproduced by the finite-width correction; this boundary region is excluded. Comparing the runs at fixed index separates a turbulent attractor, $\alpha < 0.355$, where they collapse in units of one large-scale length, from a stochastization stage still carrying the grid’s inertial range. In the attractor the extrapolated index matches the odd-ensemble theory with one scale parameter, $\chi^2/\text{dof} = 1.6$. The theory also reproduces the whole index curve of 4096^3 direct numerical simulations.

I. INTRODUCTION

“Turbulence is an essentially statistical problem of the same type as one meets in statistical mechanics, since it is the problem of distribution of energy among a very large number of degrees of freedom.”

— W. Heisenberg [1]

* amigdal@ias.edu

“To the flows observed in the long run after the influence of the initial conditions has died down there correspond certain solutions of the Navier-Stokes equations. These solutions constitute a certain manifold $\mathfrak{M} = \mathfrak{M}(\mu)$ in phase space invariant under the phase flow.”

— E. Hopf [2, p. 305]

The loop-space approach [3–5] reformulates the statistical evolution of freely decaying incompressible turbulence in terms of the circulation loop functional

$$\Psi(C, t) = \left\langle \exp \left(\frac{i}{\nu} \oint_C v_\alpha(x, t) dx_\alpha \right) \right\rangle, \quad (1)$$

the fluid analogue of a Wilson loop, see the reviews [6, 7]. In the frame moving with the fluid, the Navier–Stokes statistical evolution becomes a linear diffusion equation in loop space. A functional Fourier transform maps it to the dual momentum loop $P_\alpha(\theta, t)$, a one-dimensional field on the ordering circle. The universal decaying solution of the momentum-loop equation has the form

$$P_\mu(\theta, t) = \frac{f_\mu(\theta)}{\sqrt{2\nu(t + t_0)}}, \quad \bar{f}(\Delta f^2 - 1) = (\Delta f \cdot \bar{f}) \Delta f, \quad (2)$$

with $\bar{f}$ and Δf the mean and the jump of f at the discretization points $\theta_k = 2\pi k/N$. The nontrivial solutions, with finite vorticity, require

$$f(\theta_k^+)^2 = f(\theta_k^-)^2, \quad (f(\theta_k^+) - f(\theta_k^-))^2 = 1 : \quad (3)$$

f lies on a sphere, and consecutive points are separated by unit steps. The dynamically selected planar sector is the *Euler ensemble* [4, 5, 7], a random walk on regular star polygons,

$$\mathbf{f}_k = \hat{\Omega} \cdot \{R \cos \alpha_k, R \sin \alpha_k, \vec{0}_\perp\}, \quad R = \frac{1}{2 \sin(\beta/2)}, \quad \alpha_k = \beta \sum_{l=1}^k \sigma_l, \quad \sigma_l = \pm 1, \quad (4)$$

with $\hat{\Omega} \in SO(d)$ a global rotation. Periodicity on the ordering circle makes the turning angle rational,

$$\beta = 2\pi \frac{p}{q}, \quad (p, q) = 1, \quad \sum_{l=1}^N \sigma_l = q r, \quad r \in \mathbb{Z}, \quad (5)$$

so the turbulent state is a uniform distribution over the regular star polygons $\{q/p\}$ (Fig. 1) and over the walks on them. The continuum limit $N \rightarrow \infty$ is taken at fixed turbulent viscosity $\tilde{\nu} = \nu N^2$, which keeps the energy dissipation rate finite [4, 6]. The Euler ensemble solves the discretized loop equation exactly, as confirmed by the rigorous analysis of Brué and De Lellis [8]. The nonlinear advection term cancels identically on the compact spherical target [7].

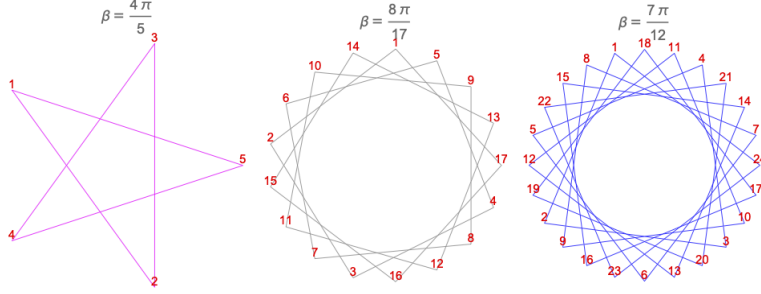


FIG. 1. Examples of the regular star polygons $\{q/p\}$ that form the discrete target space of the momentum loop. The turbulent state, the Euler ensemble, is a statistical average over random walks on all such polygons (from Ref. [6]).

Two ensembles. The continuum limit depends on the parity of N . In the odd- N ensemble both q and r must be odd, whereas in the even- N ensemble at least one of them is even. The two limits are distinct superselection sectors, $\eta = N \bmod 2$, with a different prime-2 Euler factor in the corresponding Mellin sums [7]. Only the odd ensemble is marginally Lyapunov-stable in the statistical limit: it has infinitely many decaying modes and infinitely many zero modes. The even ensemble is unstable [9]. The odd ensemble is therefore the only marginally stable turbulent attractor, and it is the one compared with experiment below.

Universal energy spectrum. The Euler ensemble gives the decaying energy spectrum in the self-similar form [5, 7]

$$E(k, t) = \frac{1}{L(t)} H(kL(t)), \quad L(t) = \sqrt{\tilde{\nu}(t + t_0)}, \quad H(\kappa) = \int_{\epsilon - i\infty}^{\epsilon + i\infty} \frac{dp}{2\pi i} M_\eta(p) \kappa^p, \quad (6)$$

with the parity-dependent Mellin amplitude

$$M_\eta(p) = \frac{f(p) \Gamma(-p) \zeta(p + \frac{15}{2})}{(2p + 7)(2p + 17) \zeta(p + \frac{17}{2}) (1 - \eta 2^{-(p+17/2)})}, \quad \eta = N \bmod 2, \quad (7)$$

where $f(p)$ is an entire function generated by the continuum limit of the Euler-ensemble path integral. It is independent of the space dimension $d > 1$ [6]. The strip of the Mellin integral is $-7/2 < \epsilon < 0$. Besides the real poles, M_η has the Riemann wall $P_n = -8 \pm i\gamma_n$, $\zeta(\frac{1}{2} + i\gamma_n) = 0$, paired with the zeros $Z_n = -7 \pm i\gamma_n$. In the odd ensemble it also has the dyadic wall $D_m = -\frac{17}{2} + 2\pi i m / \log 2$, $m \neq 0$. When the Mellin integral (6) is evaluated by Lefschetz thimbles, these walls generate the Stokes staircase of the energy spectrum: as $\log \kappa$ grows, the thimble traps the wall poles one by one, each adding an exponentially small log-periodic correction $\kappa^{-8} \cos(\gamma_n \log \kappa + \phi_n)$ or $\kappa^{-17/2} \cos(2\pi m \log \kappa / \log 2 + \phi_m)$ [7].

This paper. Experiments do not measure $E(k, t)$ at fixed t over many decades; they measure the second-order structure function $\langle (\Delta \mathbf{v})^2 \rangle(r)$ and its local slope. Here we compute the effective index of the Euler-ensemble structure function by the same thimble method used for the spectrum [7]. We work out its Stokes structure and add the finite-size correction of a tunnel of finite width. We then compare the result with the Max Planck wind-tunnel data [10] extrapolated to infinite Reynolds number.

II. THE METHOD

A. Mellin–Barnes representation of the structure function

Multiplying (6) by the three-dimensional kernel $\sin(kr)/(kr)$ and integrating over k gives, with the scaling variable $\rho = r/L(t)$ and $p = -1 - q$,

$$G(\rho) \equiv \int_0^\infty d\kappa H(\kappa) \frac{\sin \kappa \rho}{\kappa \rho} = \int_{c-i\infty}^{c+i\infty} \frac{dq}{2\pi i} \rho^q Z_\eta(q), \quad -1 < c < 0, \quad (8)$$

$$\begin{aligned} Z_\eta(q) &= -\Gamma(-1-q) \cos \frac{\pi q}{2} M_\eta(-1-q) \\ &= -\frac{\pi}{2} \frac{\csc(\pi q/2) f(-1-q) \zeta(\frac{13}{2}-q)}{(1+q)(2q-15)(2q-5) \zeta(\frac{15}{2}-q) (1-\eta 2^{q-15/2})}, \end{aligned} \quad (9)$$

where $G(\rho)$ is the scaling function of the velocity correlation $\langle \mathbf{v}(0) \cdot \mathbf{v}(\mathbf{r}) \rangle$. The structure function $D(\rho) = 2(G(0) - G(\rho))$ is the same integral with the contour shifted across the pole at $q = 0$:

$$D(\rho) = \int_{c'-i\infty}^{c'+i\infty} \frac{dq}{2\pi i} \rho^q (-2Z_\eta(q)), \quad 0 < c' < 2, \quad \alpha(\rho) = \rho \partial_\rho \log D(\rho). \quad (10)$$

The Gamma function $\Gamma(-p)$ of the spectrum is cancelled by the Fourier kernel $\Gamma(p) \sin(\pi p/2)$, leaving $\csc(\pi q/2)$. Up to normalization, (9) is the kernel $V(p)$ of Refs. [5, 6] for $d = 3$, with the odd-ensemble factor. Its poles in q are:

- the real poles $q = -1, q = 2n$ ($n \in \mathbb{Z}$), $5/2, 11/2, 15/2$ (a double pole in the odd ensemble) and $15/2 + 2n$ ($n \geq 1$);
- the Riemann wall $7 \pm i\gamma_n$, with zeros at $6 \pm i\gamma_n$;
- the dyadic wall $15/2 \pm 2\pi i m / \log 2$.

All complex singularities lie to the *right*, at small ρ .

The entire function $f(p)$ is an integral over the polygon parameter Δ ,

$$f(p) = 20 \int_{\Delta_1}^{\Delta_2} d\Delta (1 - \Delta) C^{p-1} (AC - Bp), \quad (11)$$

with $A(\Delta), B(\Delta), C(\Delta)$ given in Ref. [5]. We evaluate it on a fixed Gauss–Legendre grid in Δ , which makes $f(p)$, $f'(p)$ and $f''(p)$ exact and cheap for every complex p . Since $C(\Delta) \in [C_{\min}, C_{\max}] = [e^{-2.915}, e^{-2.763}]$, the factor $f(-1 - q) \sim C^{-2-q}$ fixes the convergence of the residue expansions:

- For $\rho > C_{\max}$ the contour closes to the left. $D(\rho)$ is then the convergent series $D = 2G(0) - 2(a_1\rho^{-1} + \sum_{n \geq 1} b_n\rho^{-2n})$ in *real* powers, and $\alpha(\rho)$ has no oscillatory component at all.
- For $\rho < C_{\min}$ it closes to the right. $D = \sum_j c_j \rho^{q_j}$ over the right poles, and the only log-periodic terms are $\rho^{7 \pm i\gamma_n}$ and $\rho^{15/2 \pm 2\pi i m / \log 2}$.

In the crossover $C_{\min} \lesssim \rho \lesssim C_{\max}$ neither series converges, and the integral must be computed directly.

B. Lefschetz thimbles and Gauss–Hermite quadrature

Write $-2Z_\eta(q)\rho^q = e^{W(q)}$, $W(q) = S(q) + q \log \rho$. For every ρ there is one real saddle $q_0 \in (0, 2)$, $W'(q_0) = 0$, since $S'' > 0$ on the strip. As in Ref. [7], the thimble through q_0 is parametrized so that the weight decays as e^{-t^2} :

$$W(q(t)) = W(q_0) - t^2, \quad \frac{dq}{dt} = -\frac{2t}{W'(q)}, \quad q(\epsilon) = q_0 + \epsilon v_0, \quad v_0 = i\sqrt{\frac{2}{W''(q_0)}}, \quad (12)$$

with the lower branch $q(-t) = \overline{q(t)}$. The Gauss–Hermite rule with nodes and weights $\{t_j, w_j\}$ then gives

$$D_{\text{thimble}}(\rho) = \frac{e^{W(q_0)}}{\pi} \sum_{t_j > 0} w_j \operatorname{Im} g(t_j), \quad g(t) = e^{W(q(t)) - W(q_0) + t^2} q'(t). \quad (13)$$

On an exact thimble the exponential factor in g equals one. Keeping it makes (13) exact by Cauchy’s theorem even where the numerical path drifts from the thimble, and doubles as a drift diagnostic. The effective index uses the same nodes, $\rho \partial_\rho D_{\text{thimble}} = \pi^{-1} e^{W(q_0)} \sum_{t_j > 0} w_j \operatorname{Im} [q(t_j)g(t_j)]$, so no numerical differentiation is needed. The ODE (12) is integrated with the exact $S'(q)$, including f'/f .

C. Stokes phenomenon: trapped poles and zero parking

Because the complex walls of (9) lie to the right, the Stokes phenomenon of the structure function is the mirror image of that of the spectrum. A pole q_n is trapped between the vertical contour and the upper thimble when the rightward ray $\{\text{Im } q = \text{Im } q_n, \text{Re } q > \text{Re } q_n\}$ crosses the thimble an odd number of times. The orientation gives a minus sign:

$$D(\rho) = D_{\text{thimble}}(\rho) - 2 \text{Re} \sum_{n \in \text{trapped}} \underset{q=q_n}{\text{Res}} [-2Z_\eta(q)\rho^q], \quad (14)$$

with the Riemann-wall and dyadic-wall residues

$$\underset{q=7+i\gamma_n}{\text{Res}} = \frac{\pi \rho^q \csc(\pi q/2) f(-1-q) \zeta(\frac{13}{2}-q)}{(1+q)(2q-15)(2q-5)(1-2^{q-15/2})(-\zeta'(\frac{15}{2}-q))}, \quad (15)$$

$$\underset{q=15/2+2\pi i m/\log 2}{\text{Res}} = \frac{\pi \rho^q \csc(\pi q/2) f(-1-q) \zeta(\frac{13}{2}-q)}{(1+q)(2q-15)(2q-5) \zeta(\frac{15}{2}-q) (-\log 2)}, \quad (16)$$

checked against contour integrals around the poles to 10^{-12} . The thimbles fall into three regimes (Fig. 2):

1. For $\log \rho > -2.8$ the thimble bends to the left, where only real poles exist, and nothing is trapped. Near $\log \rho \approx -2.87$ it rises almost vertically and reaches the walls only at heights $\text{Im } q > 80$, where the residues are negligible.
2. For $-3.8 \leq \log \rho \leq -2.95$ the thimble *parks* on a zero $6 + i\gamma_n$ of Z_η , before reaching the Riemann wall. The contour then continues up $\text{Re } q = 6$, contributing a tail smaller than 3×10^{-14} in relative terms. The index n of the parking zero jumps $1, 2, 3, 5, 6, 7, 8, 14, \dots$ as $\log \rho \rightarrow -2.87$: this is the zero-parking staircase of the structure function (Fig. 3).
3. For $\log \rho < -3.85$ the thimble escapes to the right below the first zero and traps the whole Riemann and dyadic walls above its crossing height. The first dyadic pole, at height $2\pi/\log 2 = 9.06$, is trapped for $\log \rho < -4.2$.

D. Results for the infinite system

With 80 Gauss–Hermite nodes, the thimble plus Stokes terms (14) reproduce the direct integration of (10) along $\text{Re } q = 1$ over $-8 \leq \log \rho \leq 8$. The relative deviation is at most

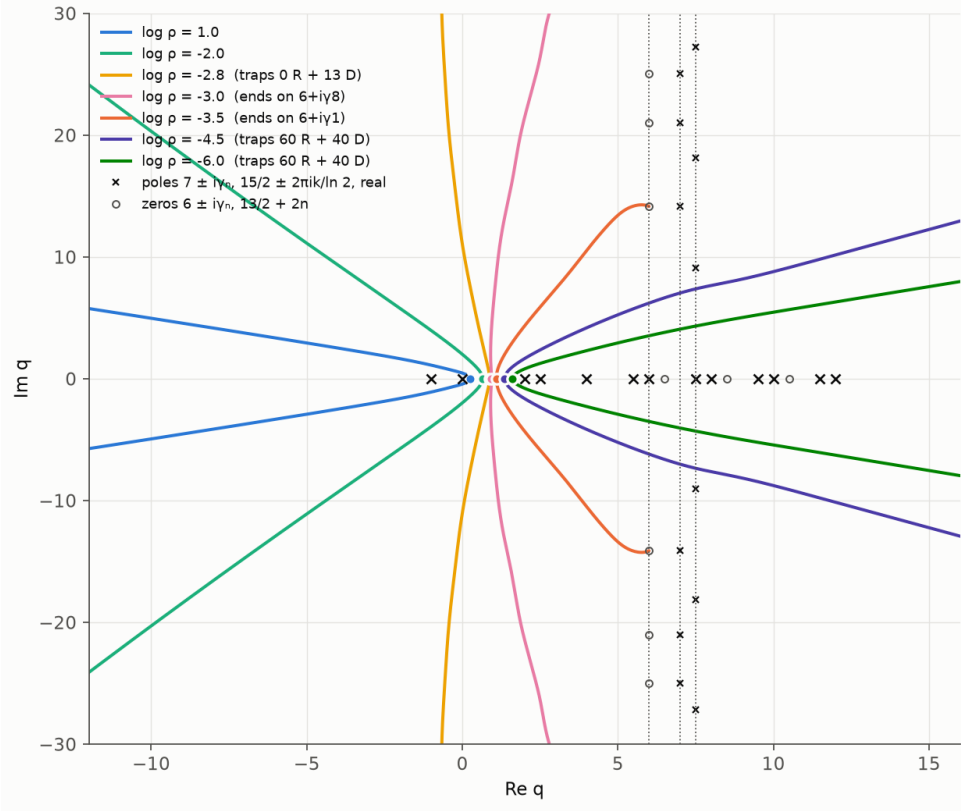


FIG. 2. Lefschetz thimbles of $\rho^q(-2Z_\eta(q))$ (odd ensemble) through the real saddles $q_0 \in (0, 2)$, for $\log \rho = 1, -2, -2.8, -3, -3.5, -4.5, -6$. Crosses are poles: the Riemann wall $7 \pm i\gamma_n$, the dyadic wall $15/2 \pm 2\pi i m / \log 2$, and the real poles. Circles are the zeros $6 \pm i\gamma_n$ and $13/2 + 2n$. Dotted lines mark $\text{Re } q = 6, 7, 15/2$.

5×10^{-10} for D , and the absolute deviation at most 4×10^{-10} for α (Fig. 4). Apart from the tiny log-periodic corrections discussed below, the effective index falls monotonically from the dissipative value $\alpha \rightarrow 2$ at $\rho \rightarrow 0$ to $\alpha \rightarrow 0$ at $\rho \rightarrow \infty$, where $\alpha = a_1/(G(0)\rho) + O(\rho^{-2})$. There is no inertial plateau: the whole crossover takes place within $-5 \lesssim \log \rho \lesssim 0$.

Oscillations of the structure function and of the spectrum. The log-periodic part of α is the sum of all wall residues. It is shown in Fig. 5, together with the log-periodic part of the spectral index $d \log H / d \log \kappa$ computed from the same walls of (7). The dyadic wall dominates both, with period $\log 2$. Where every wall pole is trapped, the thimble's Stokes terms reproduce the full wall sum to 10^{-12} . The oscillations of α are at most 1.4×10^{-6} (at $\log \rho = -3$, next to the edge of convergence) and fall as $\rho^{5.5}$: they reach 2×10^{-9} at $\log \rho = -4$ and 8×10^{-11} at $\log \rho = -4.5$. At the same distance from the convergence edge,

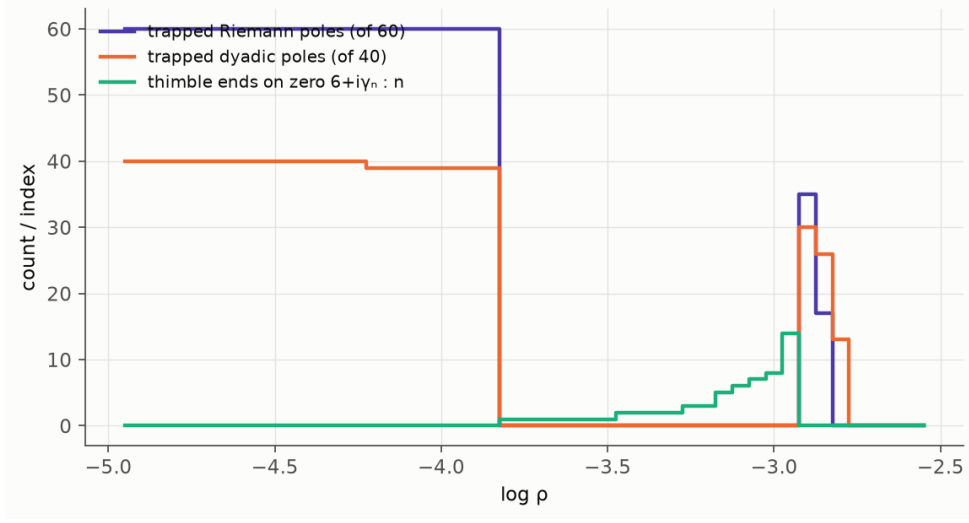


FIG. 3. Stokes staircases of the structure function: the number of trapped Riemann poles (of 60) and dyadic poles (of 40), and the index n of the zero $6 + i\gamma_n$ on which the thimble parks.

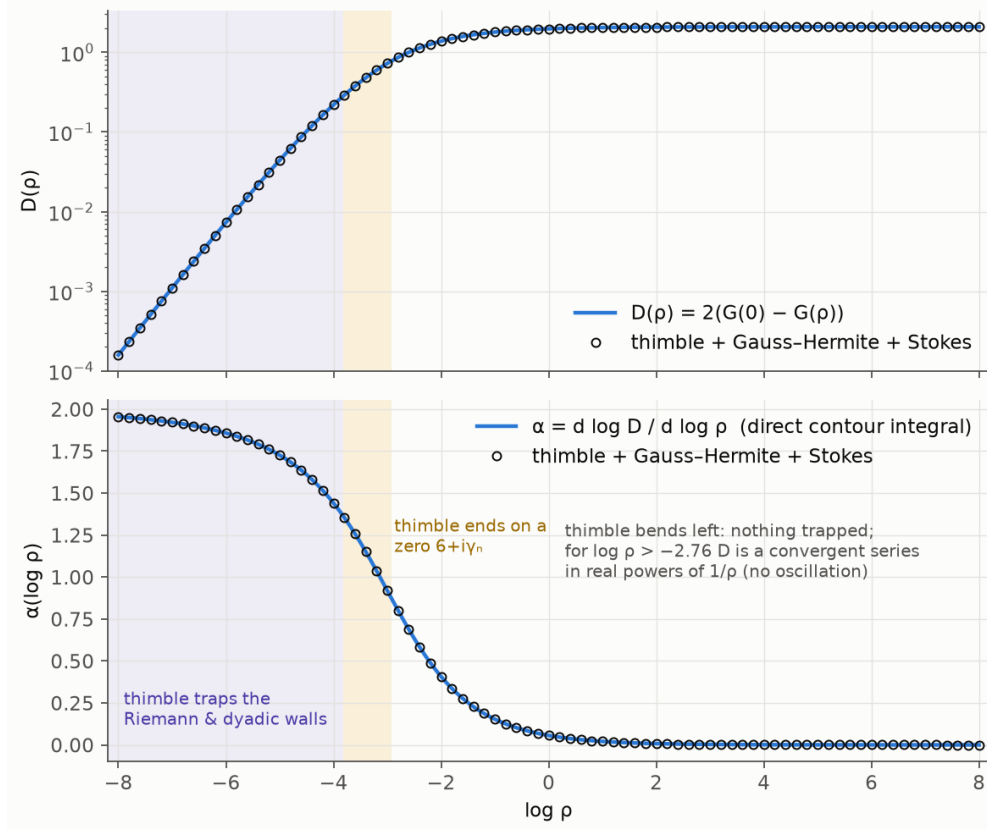


FIG. 4. The structure function $D(\rho)$ and the effective index $\alpha(\rho)$ of the odd Euler ensemble (lines: direct contour integration; circles: thimbles with Gauss-Hermite quadrature and Stokes terms). The shaded bands mark the zero-parking (orange) and wall-trapping (violet) regimes.

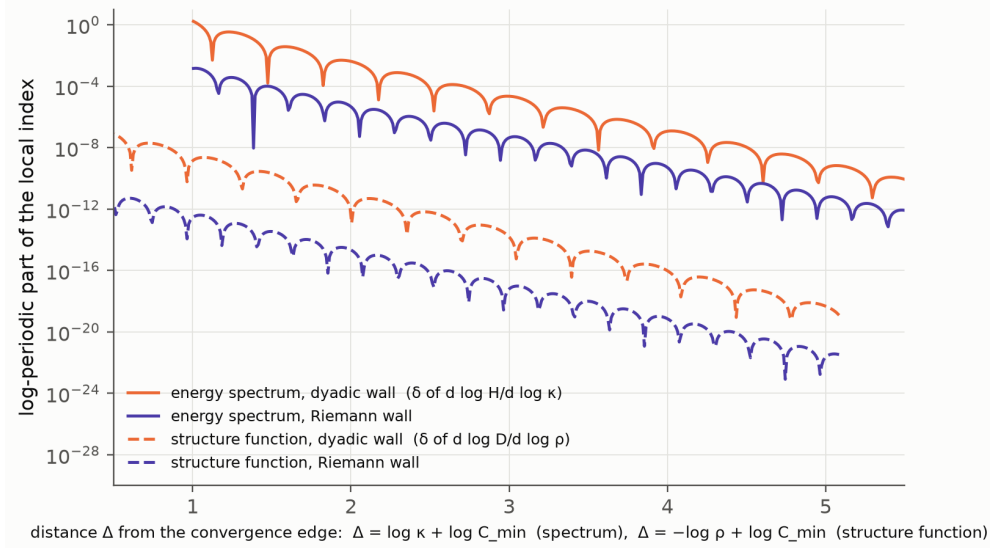


FIG. 5. Log-periodic (wall-pole) part of the local index: energy spectrum, $\delta(d \log H/d \log \kappa)$ (solid), against structure function, $\delta\alpha$ (dashed), for the dyadic and Riemann walls of the odd ensemble. The horizontal axis is the distance from the edge of convergence of the corresponding residue series, $\log \kappa > -\log C_{\min}$ for the spectrum and $\log \rho < \log C_{\min}$ for the structure function.

the oscillations of the spectral index are eight to nine orders of magnitude larger (Fig. 5). The reason is the factor $\Gamma(-p)$ in $M_\eta(p)$. At the walls its power growth $|p|^{-\text{Re } p - 1/2}$ compensates the exponential decay $e^{-\pi|\text{Im } p|/2}$, for example $|\Gamma(\frac{17}{2} - 2\pi i/\log 2)| = 1.8 \times 10^2$. In coordinate space this Gamma function is cancelled by the Fourier kernel, and only $|\csc(\pi q/2)| = 1.3 \times 10^{-6}$ remains at the first dyadic pole ($|\Gamma(8 - i\gamma_1)| = 0.33$ against $|\csc| = 4.6 \times 10^{-10}$ at the first Riemann pole). The number theory of the Euler ensemble is thus clearly visible in the spectrum [7] (Fig. 6). The structure function has the same log-periodic oscillations, with the same periods: $\log 2$ in $\log \rho$ for the dyadic wall and $2\pi/\gamma_n$ for the Riemann wall. They appear only at small separations, $\rho < C_{\min}$, the Fourier image of the large- κ tail of the spectrum, and with amplitude at most 1.4×10^{-6} . At large separations, $\rho > C_{\max}$, they are absent.

E. Finite-size correction

The Euler ensemble describes homogeneous turbulence in infinite space. In a system of finite width W the modes $k \lesssim \pi/W$ are absent or non-universal. In the scaling variables

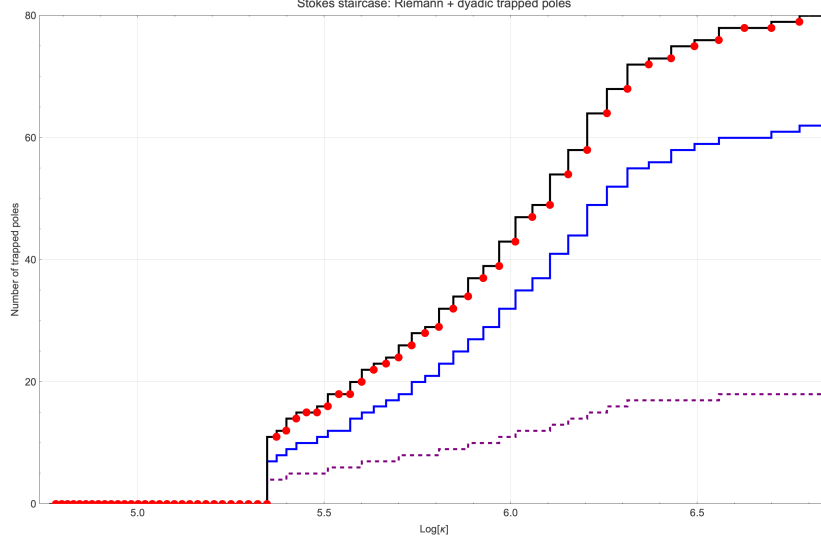


FIG. 6. Stokes staircase of the energy spectrum in the odd Euler ensemble, from Ref. [7]: the total number of trapped poles (black, dots), Riemann-wall poles (blue) and dyadic-wall poles (dashed) as functions of $\xi = \log \kappa$. Each trapped pole adds a log-periodic correction to $H(\kappa)$.

this is a cutoff $\kappa > \kappa_{\min} = \pi L(t)/W$, and the structure function becomes

$$D_W(\rho) = 2 \int_{\kappa_{\min}}^{\infty} d\kappa \left(1 - \frac{\sin \kappa \rho}{\kappa \rho}\right) H(\kappa) = D(\rho) - 2 \int_0^{\kappa_{\min}} d\kappa \left(1 - \frac{\sin \kappa \rho}{\kappa \rho}\right) H(\kappa). \quad (17)$$

The infinite-system $D(\rho)$ is the thimble result. The subtracted infrared part is computed from the small- κ expansion of H . Closing the contour of (6) to the right, only the poles of $\Gamma(-p)$ contribute, and

$$H(\kappa) = \sum_{n=0}^{\infty} \frac{(-\kappa)^n}{n!} A_n, \quad A_n = \frac{f(n) \zeta\left(n + \frac{15}{2}\right)}{(2n+7)(2n+17) \zeta\left(n + \frac{17}{2}\right) (1 - \eta 2^{-(n+17/2)})}. \quad (18)$$

Since $f(n) \sim C^n$, this is an entire function of κ . It reproduces the Mellin integral to 2×10^{-14} for $\kappa \leq 40$, and $\frac{\pi}{2} H(0)$ equals the large- ρ coefficient a_1 of $G(\rho)$. The index $\alpha_W = \rho \partial_\rho \log D_W$ uses $\rho \partial_\rho (1 - \sin x/x) = -\cos x + \sin x/x$. The lower endpoint of (17) leaves a term $\propto H(\kappa_{\min}) \cos(\kappa_{\min} \rho) / (\kappa_{\min} \rho^2)$. As a result α_W oscillates about zero with period $2\pi/\kappa_{\min} \approx 2W/L$ in ρ , periodic rather than log-periodic, and first dips below zero near $\rho \approx W/L$ (Fig. 7). These are the only oscillations of the index at large separations. An earlier evaluation of the index with the spectrum integrated over $0.1 < \kappa < 1000$ [11] found exactly such oscillations at large ρ ; they are the finite-size oscillations of (17) with $\kappa_{\min} = 0.1$.

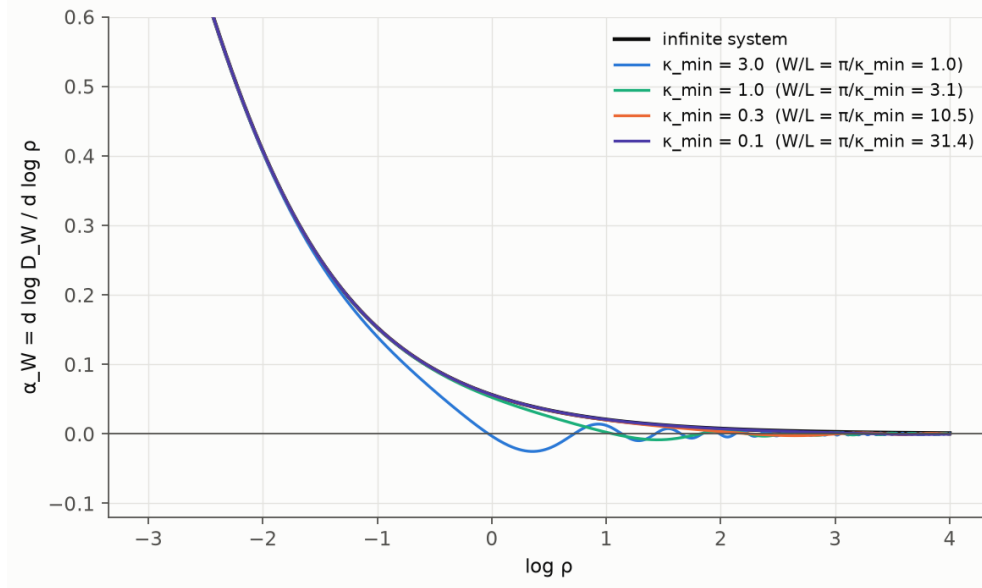


FIG. 7. The effective index of the finite system, Eq. (17), for several infrared cutoffs $\kappa_{\min} = \pi L/W$, compared with the infinite system.

III. WIND TUNNEL AND DNS DATA

A. The Max Planck measurements

We use the second-order structure functions $S_2(r)$ measured in decaying turbulence in the Max Planck Variable Density Turbulence Tunnel by Küchler, Bewley and Bodenschatz [10], for 11 runs with $Re_\lambda = 413\text{--}5779$. These are the full data sets behind Ref. [10], provided by the authors. The measured index is $\alpha_{\text{exp}} = d \log S_2 / d \log r$, computed by centered differences. The unknown normalization of S_2 drops out of α .

In grid turbulence the decay time is the distance from the grid, $t = z/\bar{v}_z$. The theory's variable is $\rho = r/\ell_D$, with diffusion length $\ell_D = \sqrt{\tilde{\nu}(t + t_0)}$. The index depends only on ρ , so a single parameter per run, $s = \log(\ell_D/\eta)$, converts r/η to ρ . The files also give u' and the Kolmogorov scale in physical units, assuming SI units for S_2 and ϵ . The viscosity inferred from Re_λ and from the dissipation range agree to 1–20%, and η ranges from $133\ \mu\text{m}$ to $8\ \mu\text{m}$. The integral scale $L = \int_0^{r_0} R dr$ of each run, with $R = 1 - S_2/2u'^2$, is 0.1–0.4 m. The fitted diffusion length is $\ell_D \approx (7\text{--}9)L$ in every run.

Each run's large- r tail, the turbulent attractor $\alpha_{\text{exp}} < 0.355$ of Sec. IIID (beyond the inertial plateau, which the theory does not contain), is fitted by $\alpha(\rho)$ with rms deviation

0.014–0.045. At the largest separations several runs show small oscillations of α_{exp} , with dips to $-0.03 \dots -0.1$ (Fig. 8). The infinite-system index cannot produce them: at these separations, $\rho > C_{\text{max}}$, it is positive and monotone, with no oscillations at all. The finite-width correction (17) reproduces them. With $\kappa_{\text{min}} = \pi \ell_D / W$ fitted per run, the rms deviation drops by up to a factor of 2.5. The implied physical width is $W = 2.25\text{--}2.6$ m for five of the seven runs with $Re_\lambda \leq 2398$, of the order of the largest measured separations (2.3–3.6 m). A single $W = 2.34$ m for these seven runs lowers their pooled rms deviation from 0.035 to 0.023. The runs with $Re_\lambda \geq 3070$ show no dip and do not constrain κ_{min} . We conclude that the observed oscillations are finite-size effects. Their amplitude does not exceed the statistical scatter of the data at the largest r , and they are not properties of decaying turbulence in infinite space.

The walls thus define a separate region of the data, the *boundary-effects region*. A separation longer than the width of the flow, $r > W$, cannot probe isotropic turbulence, and the whole flow would feel the walls once its integral scale reached the width, $L(t) \gtrsim W$. The second case does not occur here: $L \leq 0.18 W$ in every run. The first affects the last 0–19% of the points of each tail, where the dips are. These points, $r > W = 2.34$ m, are used only in this section, to find W , and are excluded from the comparison with the theory in Secs. III B and III D.

B. Extrapolation to infinite Reynolds number

The theory describes the limit $Re_\lambda \rightarrow \infty$. To compare with it we extrapolate the data, not the theory. Each run is put on the data-defined scale $x = \log(r/L)$, and at fixed x we fit $\alpha_{\text{exp}}(x, Re_\lambda) = \alpha_\infty(x) + c(x)/Re_\lambda$ over the runs with $Re_\lambda \geq 1046$, leaving out the boundary-effects region $r > W$ of each run; the three runs with $Re_\lambda < 10^3$ are decayed turbulence, too weak for the asymptotic theory (Sec. III D). The theory $\alpha(\rho)$ with $\log \rho = x - s$ is then fitted to the extrapolated tail $\alpha_\infty(x) < 0.355$, with weights $1/\sigma^2$ from the extrapolation. The result is shown in Fig. 9. A single parameter, $s = \log(\ell_D/L) = 1.953 \pm 0.017$ ($\ell_D/L = 7.05 \pm 0.12$), gives $\chi^2/\text{dof} = 1.6$ and a weighted rms deviation of 0.013, with residuals within ± 0.03 over the whole tail. The finite-width term is not needed: fitted, it goes to $\kappa_{\text{min}} \rightarrow 0$. Keeping the boundary-effects region in the extrapolation changes the scale little, $s = 1.956 \pm 0.021$, but raises χ^2/dof to 2.4; the two largest residuals of that fit, -0.16 and -0.13 at $\log(r/L) = 2.5\text{--}$

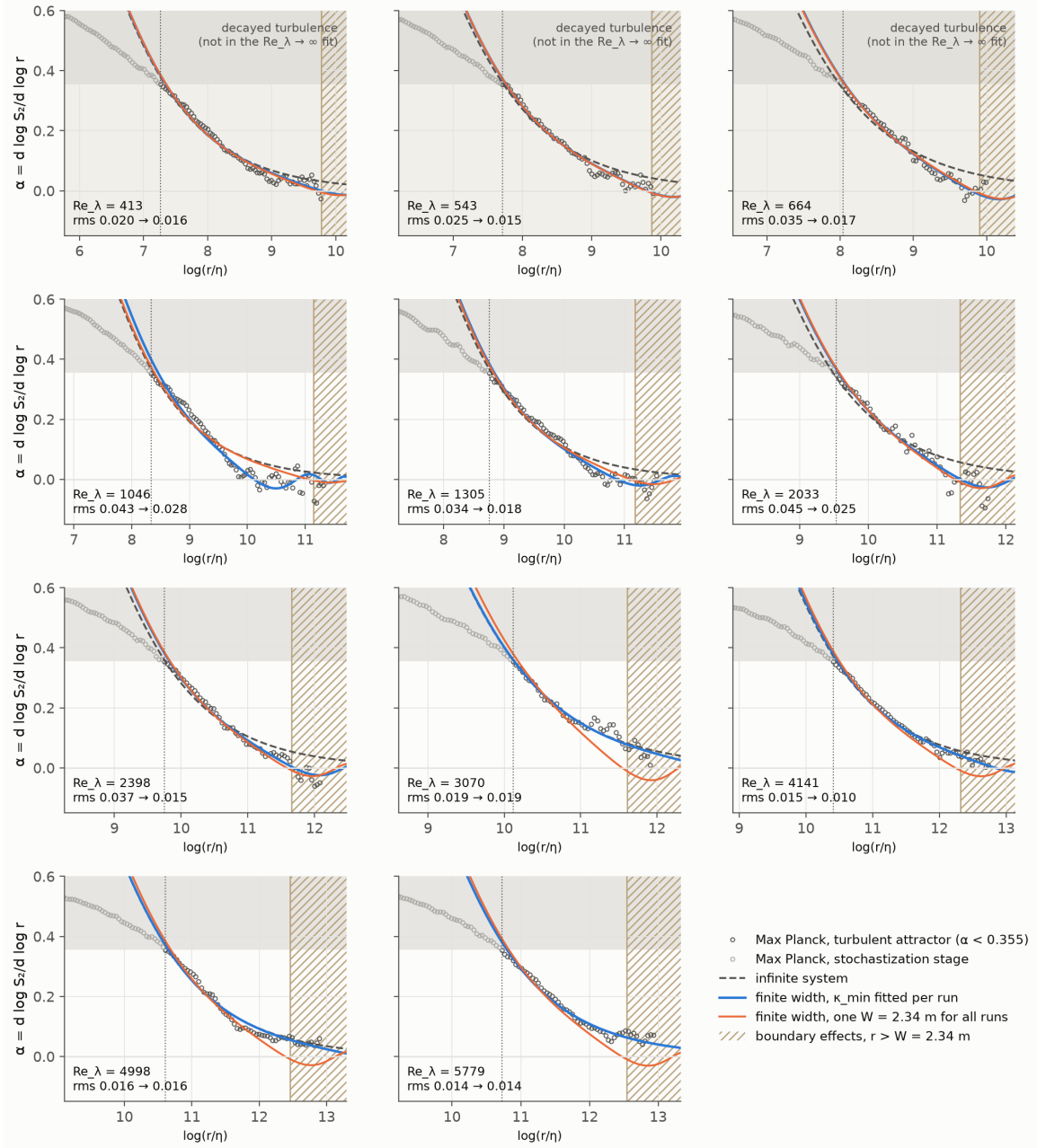


FIG. 8. Large- r index of the Max Planck runs [10] (circles) against the infinite-system theory (dashed), the finite-width theory with $\kappa_{\min}$ fitted per run (blue), and one physical width $W = 2.34$ m for all runs, $\kappa_{\min} = \pi\eta e^s/W$ (orange). The dotted line marks the start of the fitted tail. Points in the stochastization stage ($\alpha \geq 0.355$, grey band) are drawn in light grey; the tinted panels are the decayed-turbulence runs, $Re_\lambda < 10^3$. The hatched band is the boundary-effects region, $r > W$. The labels give the rms deviation without and with the per-run finite-width correction.

runs used	$s = \log(\ell_D/L)$	χ^2/dof	weighted rms of α
all 11 ($Re_\lambda \geq 413$)	2.072 ± 0.030	4.3	0.022
$Re_\lambda \geq 1046$ (adopted)	1.953 ± 0.017	1.6	0.013
$Re_\lambda \geq 1305$	1.937 ± 0.020	1.2	0.014
$Re_\lambda \geq 2033$	1.844 ± 0.012	1.3	0.012
$Re_\lambda \geq 2398$	1.811 ± 0.015	1.4	0.016

TABLE I. Fit of the odd-ensemble index to the Max Planck data extrapolated to $Re_\lambda \rightarrow \infty$ in $1/Re_\lambda$, for different sets of runs, without the boundary-effects region $r > W$. The first row includes the three decayed-turbulence runs.

2.6, come from separations beyond W . The negative dips of the individual runs therefore do not enter the comparison.

Table I shows the stability of the result under the choice of runs. The fit quality is stable, $\chi^2/\text{dof} = 1.2\text{--}1.6$, as long as the decayed runs are left out. The scale ratio ℓ_D/L drifts from 7.9 to 6.1 as the lower-Reynolds runs are dropped, more than its statistical error, so part of the Reynolds dependence of the integral scale is not captured by the $1/Re_\lambda$ term. The even ensemble gives an index within 0.0035 of the odd one on this range and the same χ^2 (1.575 against 1.577), so the tail does not resolve the parity. It is fully consistent with the odd ensemble, the only stable one.

C. Direct numerical simulation

The same theoretical index has been compared with the 4096³ DNS of freely decaying turbulence by Rodhiya and Sreenivasan [12], where the time dependence is available directly and the scaling variable is $r/L(t)$ [6]. The parameter-free theory reproduces the whole nonlinear curve of the effective index (Fig. 10). The deviations at the largest r/L come from the finite box and from early times, when $L(t)$ is still small [6]. They are the numerical counterpart of the finite-width effect in the wind tunnel.

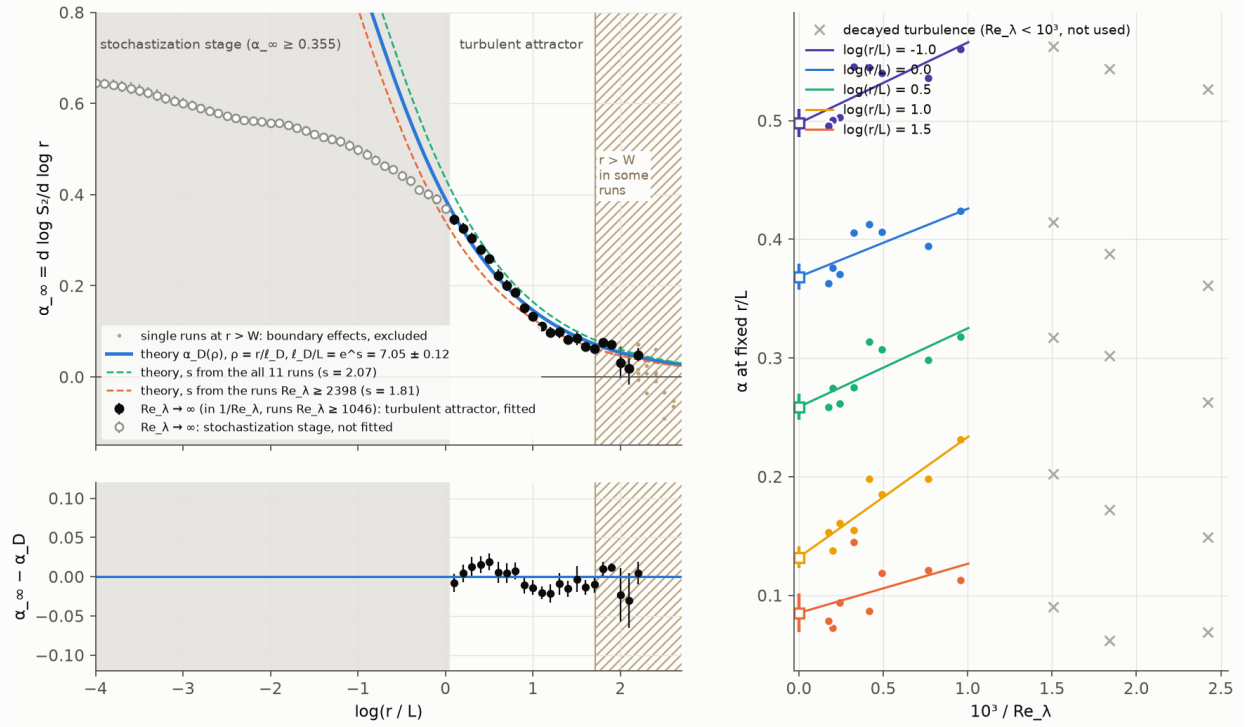


FIG. 9. Left: the Max Planck index extrapolated to $Re_\lambda \rightarrow \infty$ in $1/Re_\lambda$ (runs $Re_\lambda \geq 1046$) against the odd-ensemble theory with one scale parameter (blue). Filled circles: the turbulent attractor, $\alpha_\infty < 0.355$, to which the theory is fitted; open grey circles: the stochastization stage (grey band), not fitted. Dashed lines use the shifts fitted to the extrapolations with all runs and with $Re_\lambda \geq 2398$ only. The residuals are shown below. Right: α_{exp} against $1/Re_\lambda$ at fixed r/L ; squares are the extrapolated values, grey crosses the decayed-turbulence runs, which are not used. The hatched band starts where the first run reaches $r = W$; small tan dots are the single-run values at $r > W$ (boundary effects), which are not used either.

D. Turbulent attractor, stochastization stage and decayed turbulence

The theory describes the asymptotic state of the decay, in which the initial data have been forgotten and the index depends on r only through $\rho = r/\sqrt{\tilde{\nu}(t+t_0)}$, that is, through a single length that grows as $\sqrt{t}$. Each Max Planck run is a separate flow measured at one station, 8.3 m downstream of the active grid [10], so this scaling cannot be followed in time within a run. It can be tested across the runs at fixed index. Let r_α be the separation at which the measured index of a run crosses the level α . Where the attractor holds, $\alpha(r)$ is one function of r/ℓ , with a single length ℓ per run, and the ratio $r_\alpha/r_{0.3}$ is the same for all

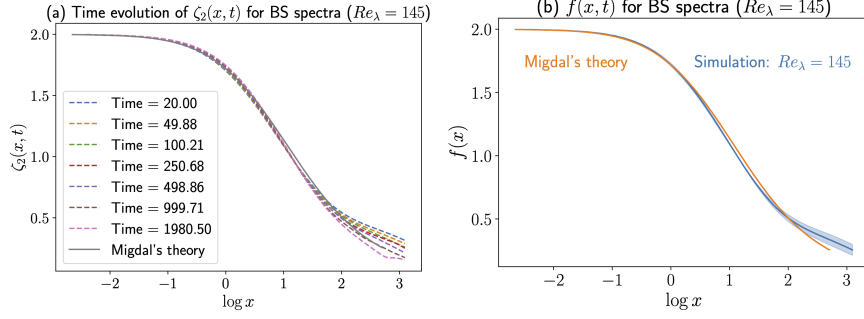


FIG. 10. Effective index $\langle r \partial_r \log(\Delta v^2) \rangle$ of the 4096^3 DNS of decaying turbulence by Rodhiya and Sreenivasan [12] (Birkhoff–Saffman initial spectrum) against the Euler-ensemble prediction (from Ref. [6]).

runs. Where it does not, r_α follows a different length.

Figure 11 shows the result for the eight runs with $Re_\lambda \geq 1046$. For $\alpha \leq 0.35$ the curves collapse tightly: the scatter of $\log(r_\alpha/r_{0.3})$ across the runs stays below 0.05, and $r_\alpha/r_{0.3}$ does not drift with the Reynolds number. Up to $\alpha = 0.5$ the scatter grows slowly, to 0.08. At $\alpha = 0.55$ it jumps to 0.21, and for $\alpha \geq 0.8$ it exceeds 0.8; there r_α follows the Kolmogorov length instead, $|d \log(r_\alpha/\eta)/d \log Re_\lambda| \leq 0.1$, as in the Reynolds-independent inertial range found by K  chler *et al.* [10]. We define the *turbulent attractor* as the region of tight collapse, $\alpha < 0.355$, roughly $L \lesssim r \leq W$. There the index follows a single large-scale length, and the theory agrees with it for every choice of runs in Table I. We call the region of larger α the *stochastization stage*. Its early part, $\alpha \gtrsim 0.5$, follows η : the flow there still carries the inertial range produced by the grid. Its late part, $0.36 \lesssim \alpha \leq 0.5$ or roughly $0.4L \lesssim r \lesssim L$, already follows the large-scale length approximately, but the measured index there falls more slowly than the theory: at $r = L/e$ it is $\alpha_\infty = 0.50 \pm 0.01$ against the predicted 0.89 (Fig. 9). Extending the fit of the extrapolated index into this range, to $\alpha < 0.40$, 0.45 and 0.50, raises χ^2/dof of the adopted fit to 1.9, 9.8 and 39, and already to 9.1 at $\alpha < 0.40$ for the runs with $Re_\lambda \geq 2398$. The three runs with $Re_\lambda < 10^3$ form a third category, *decayed turbulence*: their Reynolds number is too low for the asymptotic theory, and they are not used in the extrapolation of Sec. III B. Including them leaves both boundaries unchanged. The fourth region, the boundary effects at $r > W$ (Sec. III A), is left out of this test as well; all the crossings used here lie at $r < W$.

The collapse test has a limitation. All runs have nearly the same age at the measuring

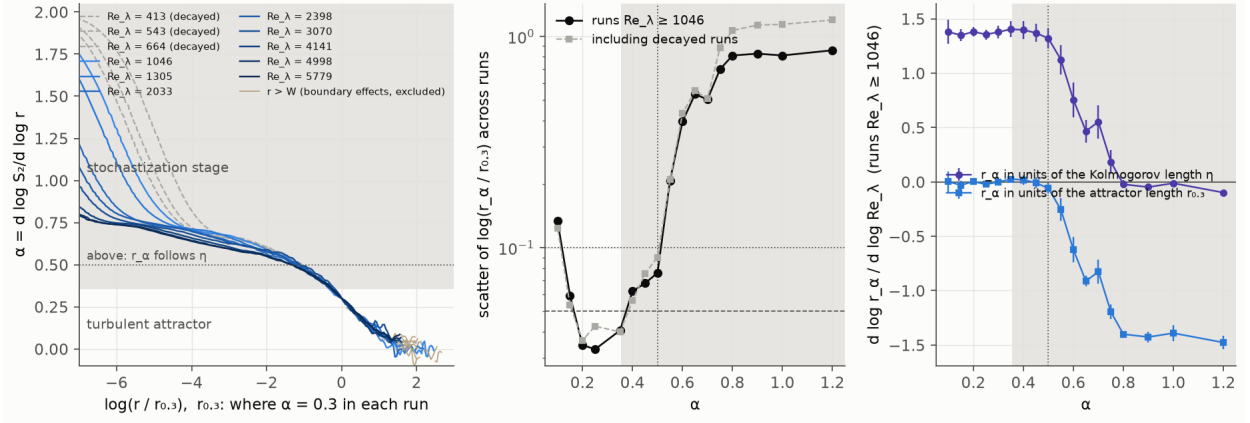


FIG. 11. Turbulent attractor and stochastization stage in the Max Planck data. Left: the measured index of each run against $\log(r/r_{0.3})$, where $r_{0.3}$ is the separation at which that run's index equals 0.3; dashed grey curves are the decayed-turbulence runs, $Re_\lambda < 10^3$. Middle: the scatter of $\log(r_\alpha/r_{0.3})$ across the runs at fixed α . Right: the Reynolds-number dependence of r_α in units of the Kolmogorov length and of $r_{0.3}$ (runs $Re_\lambda \geq 1046$). In the turbulent attractor, $\alpha < 0.355$ (white), the runs collapse tightly (scatter below 0.05, dashed line); in the stochastization stage (grey) the scatter grows, slowly up to $\alpha \approx 0.5$ (dotted lines) and steeply above, where r_α follows η . The tan ends of the curves are the boundary-effects region $r > W$, not used.

station, $\tau = u'^2/\varepsilon = 0.8\text{--}1.4$ s. A collapse across the runs therefore shows that the index follows a single large-scale length, but it cannot by itself show that the flow has reached the asymptotic state. This may be why the late stochastization stage collapses although it does not yet agree with the theory.

The finite width of the tunnel cannot produce the differences at smaller separations. The correction (17) removes only the wave numbers $\kappa < \kappa_{\min}$. Even for the largest fitted $\kappa_{\min} \approx 3.6$ it changes α by less than 10^{-3} for $\log \rho \leq -3$ and by 0.02 at $\log \rho = -1$, while the observed differences are 0.4–1.2. Nor is the stochastization stage a finite-Reynolds correction that disappears as $1/Re_\lambda$ at fixed r/L . At fixed $r/L < 1$ the measured index moves *away* from the theory as the Reynolds number grows: at $r = e^{-4}L$ it falls from 0.79 at $Re_\lambda = 1046$ to 0.66 at $Re_\lambda = 5779$, because the inertial range, which scales with η , widens toward small r/L as $L/\eta \propto Re_\lambda^{3/2}$ grows. In units of L this range extends over all $r < L$ at $Re_\lambda \approx 10^3$, where $L \approx 3 \times 10^3 \eta$, and over $r \lesssim 0.2L$ at $Re_\lambda = 5779$, where $L \approx 4 \times 10^4 \eta$. A $1/Re_\lambda$ extrapolation at fixed r/L therefore cannot remove it; it is meaningful only in the attractor

region, where the Reynolds dependence at fixed r/L is weak (Fig. 9, right).

We attribute the stochastization stage to the finite age of the wind-tunnel flow. The Euler ensemble has a single length, the diffusion length $\ell_D = \sqrt{\tilde{\nu}(t + t_0)}$, and describes the asymptotic state of the decay, $t \rightarrow \infty$, in which ℓ_D and the integral scale grow together as $\sqrt{t}$. At the measuring station the integral scale stays approximately constant along the test section [10]: the flow has not yet reached the self-similar diffusive state, and its small scales still carry the inertial range produced by the grid. Even in the DNS of Sec. III C the asymptotic decay sets in only after a long initial transient [12]; followed over very long times, the simulation reaches this state and reproduces the whole index curve. This interpretation can be tested. Since $\rho = r/\sqrt{\tilde{\nu}(t + t_0)}$, at fixed r the small- ρ part of the curve, where theory and data now diverge, is the part reached only after the longest decay times. A longer test section or a slower mean flow increases the decay time $t = z/\bar{v}_z$ at the measuring station. At fixed Reynolds number the turbulent-attractor region, and the agreement with the theory, should then extend toward smaller ρ , together with the growth $L \propto \sqrt{t}$.

IV. CONCLUSION

The steepest-descent evaluation of the Mellin–Barnes integral, with Lefschetz thimbles, Gauss–Hermite quadrature and the Stokes phenomenon, gives the Euler-ensemble structure function and its effective index with a relative accuracy better than 10^{-9} . The same Riemann and dyadic walls that produce the Stokes staircase and the log-periodic oscillations of the energy spectrum [7] are present in the structure function. There the Fourier kernel cancels the Gamma function that makes them visible in the spectrum. The index therefore has log-periodic oscillations with the same periods, but only at small separations, $\rho < C_{\min}$, and with amplitude below 1.5×10^{-6} . At large separations, where the structure function is a convergent series in real inverse powers of ρ , they vanish identically.

The small- ρ structure has a direct meaning in time. Since $\rho = r/\sqrt{\tilde{\nu}(t + t_0)}$, the limit $\rho \rightarrow 0$ is reached at small separations for fixed t or, within the validity of the Navier–Stokes description, at fixed r for $t \rightarrow \infty$. For $\rho < C_{\min}$ the structure function is the convergent sum of its right residues,

$$D(\rho) = \sum_j c_j \rho^{q_j} + \sum_n 2 \operatorname{Re} [R_n \rho^{7+i\gamma_n}] + \sum_{m \neq 0} 2 \operatorname{Re} [R'_m \rho^{15/2+2\pi i m / \log 2}], \quad (19)$$

where $q_j = 2, \frac{5}{2}, 4, \frac{11}{2}, 6, \dots$ are the real poles. At fixed r the wall terms oscillate in $\log t$ with the frequencies $\gamma_n/2$, set by the non-trivial zeros of $\zeta(s)$, and $\pi m/\log 2$, set by the prime-2 Euler factor of the odd ensemble. Their amplitude relative to the leading term decreases as $\rho^5 \propto t^{-5/2}$. This infinite set of incommensurate complex exponents makes $t = \infty$, equivalently $\rho = 0$, an essentially singular point of the structure function, exactly as $\kappa = \infty$ is for the energy spectrum [7]. Under $p = -1 - q$ the exponents $-8 \pm i\gamma_n$ and $-\frac{17}{2} \pm 2\pi i m/\log 2$ of the spectrum map one to one onto $7 \mp i\gamma_n$ and $\frac{15}{2} \mp 2\pi i m/\log 2$. The infinite-time singularity of the velocity structure function is therefore dictated by the same number theory as the spectral singularity. Only the order of the Stokes activations differs. In the spectrum the thimble captures the Riemann-wall poles one by one, at times $t_n \propto \gamma_n^3/(\tilde{\nu}k^2)$ at fixed k [7]. In the structure function, where the Fourier kernel cancels $\Gamma(-p)$, the thimble captures the whole walls at once, for $\log \rho \lesssim -3.85$. In both representations the singular structure of decaying turbulence at infinite time is encoded by the zeros of the Riemann zeta function: number theory decodes turbulent intermittency.

The small oscillations of the index observed at the largest separations in the Max Planck wind tunnel are reproduced by the finite width of the tunnel. They lie at separations beyond the width, $r > W$, which cannot probe isotropic turbulence; this boundary-effects region is excluded from the comparison with the theory, which improves the fit. The oscillations are of the order of the statistical scatter or below it. They are not a property of decaying turbulence in infinite space. The extrapolated tail of the measured index matches the odd Euler ensemble, the only stable turbulent attractor, with a single scale parameter. The 4096³ DNS reproduces the whole index curve. A comparison of the runs at fixed index separates the turbulent attractor, $\alpha < 0.355$, where the runs collapse onto one curve in units of a single large-scale length and agree with the theory, from the stochastization stage at larger α , where the collapse loosens and, above $\alpha \approx 0.5$, the index follows the Kolmogorov length and the flow still carries the inertial range produced by the grid; the runs with $Re_\lambda < 10^3$ are decayed turbulence, and the separations $r > W$ are boundary effects. In the late stochastization stage, $0.36 \lesssim \alpha \leq 0.5$, the measured index falls more slowly than the theory. The stochastization stage is not a finite-size effect, and it is not a finite-Reynolds correction that vanishes as $1/Re_\lambda$ at fixed r/L . We attribute it to the finite age of the wind-tunnel flow (Sec. III D).

The same attractor also fixes the decay of the turbulent kinetic energy, $E \propto t^{-5/4}$ [5], or

$E \propto z^{-5/4}$ in grid turbulence, where $t = z/\bar{v}_z$. This law was observed already in the classic grid-turbulence experiment of Comte-Bellot and Corrsin [13], and it agrees with the long-time DNS of Rodhiya and Sreenivasan [12], who also analyze how finite-size and initial-state corrections influence the decay exponents extracted from numerical and laboratory experiments. The energy decay and the tail of the structure function test different consequences of the same solution. Together they are one more verification of the number theory of turbulence.

Wind-tunnel experiments with a larger cross-section, reaching larger separations in units of the diffusion length, and at higher Reynolds numbers, would separate the finite-size effects from the bulk dynamics. They would also test the predicted tail $\alpha \simeq a_1/(G(0)\rho)$ over a longer range. A longer tunnel or a slower mean flow would increase the decay time $t = z/\bar{v}_z$ at the measuring station and, since small ρ at fixed r means large t , should extend the agreement toward the small- ρ part of the index, where the present data depart from the theory. Measurements at several distances from the grid would follow this approach to the asymptotic state.

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DECLARATION OF GENERATIVE AI IN THE WRITING PROCESS

The Python implementation of the computations, the analysis of the wind-tunnel data and the first draft of this manuscript were prepared with the assistance of generative AI (Claude, Anthropic), following the author’s instructions and his *Mathematica* notebooks. The author verified the scientific content and takes full responsibility for the manuscript.

CODE AND DATA AVAILABILITY

The code that produced every number and every computed figure of this paper is provided as ancillary files with the arXiv version of the paper (directory `anc`) and is maintained in the folder `Mellin` of the GitHub repository [14]; version v2.0 is archived on Zenodo [15]. It contains (i) the Python implementation used for all results: the table of $A, B, C(\Delta)$ and the function $f(p)$, the saddles, thimbles, Gauss–Hermite sums and Stokes terms of Sec. II, the direct contour integrals used as checks, the finite-width correction (17), the analysis of the wind-tunnel data of Sec. III, and the script `run_all.sh`, which regenerates Figs. 2–5 and 7 and, given the Max Planck data, Figs. 8, 9, 11 and Table I; (ii) a *Mathematica* package and driver notebook implementing the same algorithm; (iii) the scan tables and fitted parameters. The file `README.md` there lists the script behind each figure, table and quoted number. Earlier notebooks are in Refs. [11, 16, 17]. The Max Planck data of Ref. [10] were provided by the authors on request and are quoted with their permission; they are not redistributed and are available from the authors. The DNS data of Ref. [12] are available on request from their authors.

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